

accessible to all. This is a vital step on the path to creating a shared information network for safer roads. If a car around the next corner hits the brakes because there is an obstruction, that information could be used to signal to the drivers behind to slow down ahead of time, resulting

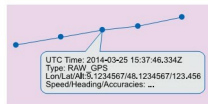


Fig. 1: Paths

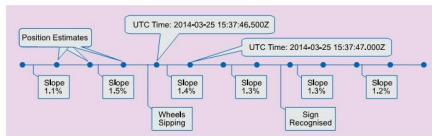


Fig. 2: Path events

in smoother, more efficient journeys and lower risk of accidents. But that can only work if all cars can speak and understand the same language.

SENSORIS

Vehicles driving on the road are equipped with a magnitude of sensors. This sensor data may be transferred over any kind of technology from the vehicle to an analytic processing backend. Between individual vehicles and the analytic processing backend, an OEM or system vendor backend may be located as a proxy.

The sensor data interface specification defines the content of sensor data messages and their encoding format(s) as these are submitted to the analytic processing backend. However, the specification may be used between other components as well. Sensor data is submitted as messages with various types of content. It is common to all kinds of submitted messages as these are

related to one or multiple locations.

Sensor data messages may be time-critical and submitted near-real-time but also may be of informational value and submitted with an acceptable delay accumulated within other data. Neither priority nor requirements on latency are part of the specification. The content and format of sensor data messages is independent of submission latency (in near-real-time or delayed).

Vehicle metadata provides information about the vehicle that is valid for the entire path. This includes vehicle type information and vehicle

level, these raw measurements must be converted into meaningful values and, hence, are a result of calculations either in the vehicle or in the OEM or system vendor backend.

The fundamental data element for rich data sensor submission is a path, which is a list of position estimates, which are ordered starting with the oldest position estimate. A path can be very short, for example, for near-real-time events that are transmitted immediately after these occur. It can be very long, for example, like an entire drive over many hours that records the vehicle trace and events for later submission.

Vehicles may have different collection policies for different types of sensor data. According to individual priorities of sensor data information, a message could be compiled and sent out the moment specific sensor data reading is detected, or events could be accumulated into one message and submitted after a given amount of time.

By way of example, this could be due to reducing computational performance in critical moments while driving due to reduction in transmission volume through mobile phone connection, or due to accumulation and, thus, referencing all detected path events onto one single path. The latter may happen if referencing of multiple path events to one single vehicle is needed and usage of transient vehicle ID is not supported or not wished.

Self-driving cars

A self-driving car is capable of sensing its environment and navigating without human input. To accomplish this task, each vehicle is outfitted with a GPS unit, an inertial navigation system and a range of sensors including laser range-finders, radar and video. The vehicle uses positional information from the GPS and inertial navigation system to localise itself and sensor data to refine its position estimate as well as to build a 3D image of its environment.